

XJCO3011 — Coursework 2: Search Engine Tool

Submission Document

Student Name: Minhao Gao

Student ID: 201691058

Module: XJCO3011 — Web Services and Web Data

Submission Date: May 2026

(A) Video Demonstration Link

Platform: Google Drive / YouTube / OneDrive (public link required)

Video Link: [TO BE FILLED AFTER RECORDING]

The video must be accessible via a public URL. Test the link in incognito mode before submitting.

Video Requirements:

- Duration: Maximum 5 minutes (strict)
- Format: MP4 or MOV
- Content: Live demo + design explanation + testing + Git history + GenAI reflection
- Must cover all 5 components per coursework brief

(B) GitHub Repository URL

Repository: <https://github.com/Bartley007/xjco3011-coursework1/tree/main>

Repository Contents:

File / Directory	Description
.gitignore	Git ignore rules
requirements.txt	Python dependencies (requests, beautifulsoup4)
README.md	Project documentation with setup/usage instructions
src/	Core source code modules
■■ crawler.py	Web crawler for quotes.toscrape.com (BeautifulSoup)
■■ indexer.py	Inverted index builder (term frequency + positions)
■■ search.py	Intersection-based multi-word search engine

■■ main.py	Interactive CLI (build/load/find/print/exit)
tests/	Unit test suite
■■ test_crawler.py	Crawler tests with mocked HTTP responses
■■ test_indexer.py	Indexer tests (tokenization, build, save/load)
■■ test_search.py	Search tests (single/multi-word, edge cases)
data/	Generated index files (index.json + docs.json)

Git Commit History (10 commits):

```
9c213ae | Apr 20 | feat: implement web crawler
b7aadf2 | Apr 21 | feat: implement inverted indexer
a643e75 | Apr 22 | feat: implement search engine
e6eaf52 | Apr 23 | feat: implement CLI main loop
c5e3054 | Apr 27 | test: add unit tests
d134d18 | Apr 28 | docs: add README
70b4873 | Apr 29 | chore: update requirements
1b83cc3 | Apr 30 | feat: integrate all modules
d4b4661 | May 01 | docs: GenAI declaration
c5d1bc9 | May 02 | docs: finalize for submission
```

(C) Index File (Inverted Index)

File Name: data/index.json + data/docs.json

Description: Serialized inverted index generated by running build command.

Index Structure:

Component	Format	Details
index.json	Dict[word → Dict[doc_id → List[position]]]	Term frequency + position tracking for ranking
docs.json	List[{text, author, tags, url}]	Full document storage (100 quotes from quotes.toscrape.com)

Note: The index file will be attached separately on Minerva or included as a download link if too large.